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INDO-EUROPEAN AND NOSTRATIC: SOME FURTHER COMMENTS (A RESPONSE TO "EXPLORING THE NOSTRATIC HYPOTHESIS")

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1. *Introduction*

In setting down some observations that reflect, at least in part, my extempore remarks as discussant during the Ypsilanti conference, I am happy to have been given the opportunity to comment on the stimulating paper "Exploring the Nostratic Hypothesis", by A. Manaster Ramer, P. Michalove, K. Baertsch, and K. Adams (hereafter "AMR et al.", for expository convenience). The title of my response recalls my 1991 paper "Indo-European and Nostratic", to which AMR et al. refer on a number of occasions; my primary goal here, then, is to address the Nostratic enterprise — in particular certain points raised by AMR et al. — from the point of view of Indo-European linguistics. In my earlier paper, I remarked (1991:31) that Nostratic theory is in many ways "an elegant and well-articulated structure", but that "with respect to the treatment of Indo-European data, it remains as yet a house of cards". Since 1991, I have been able to deepen my acquaintance with Nostratic theory, not least of all through the Ypsilanti conference itself, as well as the kindness of Dr. Manaster Ramer, with whom I have had the benefit of many discussions (mainly *per litt. electron.*). Although my overall assessment remains more or less what it was, I have been deeply impressed with much of what I have learned, and would prefer at this time to emphasize the "well-articulated structure" (a house of cards is at any rate a house of sorts), as well as, in the concluding words of my earlier paper, the "enormous potential" of Nostratic studies "to contribute to our understanding of human linguistic history". (It should thus be clear, incidentally, why I have nothing to say about §1 of AMR et al.'s paper ["Early Responses to the Theory"], beyond approving their censure of various facile rejections and misrepresentations of Nostratic theory.)

My restriction to an Indo-European focus is partly an inevitable consequence of my own training and research, but it is also related to a point of personal conviction, to be discussed in §3.1 below.

2. *Comments on "Exploring the Nostratic Hypothesis", §2 (general issues)*
In §2 of their paper ("A New Approach to Nostratic"), AMR et al. review four basic "principles that have been applied to other language families", and consider their application to Nostratic. As AMR et al. note, "these principles are well understood by scholars working in other language families"; I therefore find them essentially uncontroversial, and have little to add beyond certain qualifying remarks connected specifically with Nostratic material. I comment on each principle in turn.

2.1 *Which languages?*
AMR et al. write: "First, it is relatively unimportant at this stage whether the six languages considered in Illič-Svityč's work comprise all of Nostratic, or whether others should be added, or some dropped", since "questions about the composition of Nostratic do not invalidate the theory". Agreed. My 1991 treatment made no reference to this issue, beyond noting (1991:30, 31 n.20) the existence of some alleged cognate material from Sino-Tibetan and Austronesian — material that might or might not be accepted by all scholars working in a Nostratic framework, and that as such clearly has no bearing on the general validity of Nostratic.

2.2 *How many etymologies are needed?*
AMR et al. write: "Second, there are clearly many etyma in Illič-Svityč's work that are faulty", but "surely 50 or 100 truly convincing cases, sufficiently represented among all or most of the languages involved, would be enough for us to accept the relationship of these families", since "Other language families have been accepted with relatively few cognate sets". Agreed in principle, but with a serious qualification pertaining to Nostratic. In support of this argument, AMR et al. invoke, quite appropriately, interesting statistics drawn from Goddard's study of "Sapir's comparative method" (1986), in which Goddard reevaluated the methodology of Sapir's attempts to demonstrate a genetic relationship between Algonquian and Ritwan. AMR et al. note that while modern scholarship accepts Sapir's demonstration as correct, this is so despite Goddard's rejection of between 76% and 82% of Sapir's etymologies, and his demonstration that (as AMR et al. put it) "only 35, or 18%, of the 196 morphemic and lexical comparisons proposed by Sapir (1914) between Algonquian and Ritwan are definitely correct, and another 13, or 7%, are likely to be so". I would note, however, that Sapir's material included both "*morphemic* and lexical comparisons" [emphasis mine — BV]; indeed, one of the main points of Goddard's paper was that Sapir's *morphological* evidence was sufficiently strong to have made it puzzling as to why he felt compelled to add large numbers of relatively weak *lexical* comparisons. This observation, in turn, is directly related to the importance of morphological correspondences in demonstrating genetic relationship, as Goddard himself discussed in his classic paper (1975) on proving the Algonquian/Ritwan relationship, which I cited in this connection in my earlier treatment (1991:18). I therefore continue to be somewhat troubled by the possible absence, thus far, of convincing (at least to my mind) Nostratic etymologies involving Indo-European morphological (as opposed to lexical) material. AMR et al. offer no new examples for consideration, even noting, toward the end of their §2, two of Illič-Svityč's morphemic comparisons that "cannot be accepted today" (a Nostratic "causative-desiderative" morpheme,[1] and alleged Turkic and Kartvelian congeners to the Germanic dental preterite). As I suggested earlier (1991:16), the problem is exacerbated by the relative brevity of many of the morphemes under consideration.

As an example of what I judged to be a highly problematic use of Indo-European morphological material in support of a Nostratic etymology, I discussed (1991:18f.) certain problems and indeterminacies surrounding the Indo-European material Illič-Svityč had adduced in support of the Proto-Nostratic adjective suffix *-lʌ (1976:20, #253). I. Hegedűs has since sought, in her words, to "enhance the plausibility" of this and other Nostratic etymologies, in a paper on "Reconstructing Nostratic Morphology" (1992, see 1992:35-7 on PN *-lʌ). For the Indo-European part of the equation, Hegedűs provides a blanket reconstruction for a morpheme "*-l-" (1992:35), which she describes (following Illič-Svityč) as a "suffix of denominative and deverbal adjectives". Her comments on the Indo-European material read, in full, as follows: "INDO-EUROPEAN. The diminutive meaning is attested in the IE daughter languages, e.g.: Goth. *barnilō* (*barn* 'child'), Lat. *rēgulus* (*rex* 'king'), OInd. *śiśūlas* (*śiśuš*[sic] 'child'), Gr. παχυλως [sic] 'somewhat fat' (παχύς 'fat')."[2] These examples supplement those provided by Illič-Svityč, who included no material from Greek and Indic, and who said nothing specific about the diminutive function; this latter, moreover, is evidently meant to help align the IE data more closely with material elsewhere, since Hegedűs specifically points to the diminutive function of corresponding *-l- formants in Kartvelian and Uralic. Nevertheless, these IE data in no way "enhance the plausibility" of the etymology in question, even apart from certain problems

[1] Partly based on IE material, on which see recently Dolgopolsky 1992a:293ff.
[2] A footnote directly following observes that "As a deverbal adjective formant it is reconstructable for PGmc. as *-alo-/-ulo- ".

with some of the forms themselves.[3] While a diminutive function is often loosely reconstructed for IE *-lo-, this may have involved, more specifically, endearing and hypocoristic (partly also patronymic) formations restricted to naming-expressions (and other terms referring to humans, like Goth. *barnilō*): thus Skt. *Aśvala-*, Gk. Κρατύλος, Goth. *Wulfila* etc. (see Leumann 1977:310f. for discussion and references). Nor is there any clear sense in which this behavior of IE *-lo- (with rather diverse einzelsprachlich avatars: thus also *-elo- in Italic and Germanic, etc.) reflects the "same" formant as the verbal adjective *-lo- (as in Armenian, Slavic, and Tocharian [participles, infinitives and verbal nouns/adjectives], the agentive *-lo- of Lat. *legulus* 'gatherer', etc.), not to mention various other nominal and adjectival *-lo- formations ("complex Caland" *-u-lo- alternating with *-u-ro-, etc.).[4] With semantic and formal characterizations as broad and crude as those used by Illič-Svityč and Hegedűs, we might just as persuasively compare the Anatolian "pronominal -l-" (emphatic forms Hitt. *ukel* 'I myself', *apašel* 'he himself' etc., genitives Hitt. *apēl* 'eius, illius' = Lyd. *bil*, Hitt. *kēl* 'huius' etc.), which according to Hamp (1982) may underlie certain IE *-lo- formations, with the similar Proto-Algonquian pronominal marker (or "pseudo-root", as per Bloomfield 1946: 116, §88) *-i:l- (e.g. *ni:la 'I' [i.e. *n-i:l-a; Fox *ni:na*, Cree *niya*, Menomini *nenah*], *wi:la 'he, it' [Cree *wiya*, Men. *wenah*], etc.). I suspect, however, that few except "Proto-World" enthusiasts will wish to argue on such a basis for a "Proto-Indo-Algonquian".

Nor does Hegedűs provide convincing support from Indo-European for the other two morphemes she discusses. As for Proto-Nostratic *na ("formative of verbal and nominal relative constructions", Hegedűs 1992:37ff.), Illič-Svityč compared IE *-no-, which he defined once again as a "deverbal and denominal adjective suffix", but he did not actually cite any relevant forms, given the chart format of his presentation (1971:16). Hegedűs, however, adduces (1992:39) both the typically full-grade type of deverbal noun in *-no-, as in *swép-no-/ *swóp-no- 'sleep',[5] as well as the typically zero-grade type of deverbal adjective in *-no-, as in *pl̥h₁-nó- 'full'.[6] Again, no explicit connection is made between these two quite distinct formations, nor is it clear how such forms would continue primordial "verbal and nominal relative constructions".[7] Still more wildly varied are the IE t-formations Hegedűs invokes (1992:42) in support of a Proto-Nostratic morpheme *t, for which Illič-Svityč did not argue as such, but which Hegedűs describes as a "suffix or prefix forming deverbal or denominal nouns, mainly abstracta" (1992:41). I stress, once again, that I refrain here from evaluating non-Indo-European evidence, and that my comments do not therefore invalidate the Nostratic etymology per se (more below on the general problem raised by this point); but I fail to see how Indo-European formants as disparate as *-tó-[8], *-to-[9], *-ti-[10], *-et-/*-e-t-[11], and *-oto-/*-eto-[12] can justify the reconstruction of a "PIE *-t-" (thus Hegedűs 1992:42), which might then be used to support a (monophonematic!) Nostratic "etymology". (Nor does the almost indiscriminate use of such an Indo-European hodgepodge bolster one's confidence in the handling of the non-Indo-European data.)

[3] The Vedic hapax *śiśūla-* (RV 10.78.6) probably does not mean 'small child', is almost certainly a compound with laryngeal-initial second member (note the length of the second syllable, vs. e.g. *śiśuká-* 'child'!), and therefore does not have a simple l-suffix at all (see Mayrhofer 1996:642, with further references); Gk. -υλο- (and -υλλο-) forms with clear diminutive force are attested relatively late, and may reflect a secondary development in Greek — even the adj. *παχυλο- underlying the adv. παχυλῶς (first attested in Aristotle) need not be interpreted as a diminutive, but could well reflect a "secondary complex Caland" formation in *-u-lo- (see in general Nussbaum 1976), formally comparable to Ved. *bahulá-* 'thick, broad' (Mayrhofer 1993:220f.).

[4] The IE *-l- formations are far from having been definitively interpreted, as I noted at 1991:19 with reference to the preliminary treatment of Solta (1970); for an interesting collection of material from Slavic, see recently Pavlova 1994, including some forms that may show relatively old *-l- formations (e.g. Pol. *gomóla* 'lump' and other Slavic forms beside Li. *gãmalas* 'heap (of snow); lump, piece (of bread)', Pavlova 1994:104).

[5] The zero grade variant *sup-no-, as apparently in Greek and Slavic, can be secondary, as I discuss elsewhere (1998a).

[6] The full grade of Lat. *plēnus* is probably secondary (see e.g. Schrijver 1991:184), and Arm. *li*, pace Hegedűs, need not contain a nasal suffix at all (see e.g. Schmitt 1981:51).

[7] Even if the latter feature were taken to be an automatic consequence of defining these formations as essentially "adjectival", the *swép-no- type is at best characterizable as forming *substantivized* adjectives (thus also *déh₃-no- 'gift', *wés-no-/*wós-no- 'price', *wégh-no-/*wógh-no- 'wagon, conveyance', and a number of others, on which see recently Barton 1993:554), as distinct from the verbal adjective *-nó- with zero grade.

[8] As in the familiar pret. participles with zero grade root.

[9] I.e. root-stressed forms with full grade (especially o-grade), forming mainly abstract and concrete nouns of various types — Hegedűs cites Gk. πλοῦτος 'wealth'; also e.g. Lat. *hortus* 'garden, enclosure', OPr. *dalptan* 'chisel'. (On the formation of the latter see Toporov 1975:293. Polomé [1986:662f.] seems to take the root underlying OPr. *dalptan* and its Germanic cognates as non-Indo-European, but the formation of the OPr. term is in any case regular.)

[10] Mainly action nouns, as in the familiar type with zero grade root.

[11] E.g. Ved. *sravát-* 'stream' (agent noun?), the sole form of this type cited by Hegedűs; cf. also the type Lat. *teget-* 'mat', and the somewhat different type Hom. ἀργετ- (oblique stem of ἀργής 'bright').

[12] The profile of these formants (which actually subsume several distinct subtypes) is not entirely clear, as I discuss in detail elsewhere (1998b).

In conclusion, I would qualify the otherwise quite reasonable statement of AMR et al. that "The Nostratic hypothesis ... can easily afford to lose many, even a majority of the etymologies and still be correct" by adding a rider, roughly to the following effect: "especially if a core of solid morphological correspondences can be demonstrated". Leaving aside for the moment the personal pronouns (which pose special problems, to be raised in the next section), I am thus far not convinced of the validity of any reconstructed Proto-Nostratic morpheme that depends to any significant extent on Indo-European material: in all cases known to me, the IE material itself is too vaguely described, incorporates subtypes that are too disparate, or is too susceptible of being explained via universal or typological tendencies (cf. next section) or by the workings of chance (in the case of loosely-defined mono- or biphonematic morphemes) to be useful in support of any such reconstruction.[13]

2.3 *On the rate of linguistic change*

AMR et al. write: "Third, we must be wary of blanket generalizations about how much linguistic change we can assume over a given amount of time", since "different languages undergo change at differing rates". Agreed. As with the item in §2.1 above, this point did not figure in my 1991 treatment, except in the following indirect way: I referred to claims about the relative "stability" of certain morphemes over time, observing (1991:16-17) that despite such claims, one must reckon with the effects of behavior similar to phonaesthetic patterning even among grammatical morphemes.[14] It is precisely for this reason that I prefer to withhold judgment as to the validity of the reconstructions proposed for Nostratic personal pronouns (e.g. Illič-Svityč 1976:52, #289 [PN *mä, 1 pl. incl.]; 1976:63, #299 [PN *mi, 1 sg.]), despite the confidence often placed in these.[15] In a pioneering cross-linguistic study of the phonological composition of personal pronouns, based on material from 62 languages, Gordon (1995) highlighted certain recurrent features of personal pronouns that might best be considered in typological terms, thereby complicating considerably the evaluation of genetic hypotheses — especially long-range ones — involving such material. In general, Gordon's results have been

[13] On the problems associated with several attempts to elucidate "deep" morphological comparisons involving Indo-European material, see my earlier discussion (1991:19-24).

[14] The example under discussion involved widespread evidence for a "negative *ma*" (cf. PN prohib. particle *mă, Illič-Svityč 1976:56, #290), with which I compared remarks by Jespersen on universal tendencies for *m-* in adversative conjunctions.

[15] Thus AMR et al. write: "It may be that in some years we will say that the study of Nostratic began with, say, the personal pronouns, ..."; on some details of the reconstruction of the Nostratic personal pronouns, see also Manaster Ramer 1993:216ff.

confirmed by the more extensive study of Nichols and Peterson (1996), based on material from 173 languages.[16]

2.4 *On borrowing*

AMR et al. write: "Fourth, a consideration of Nostratic must make further use of borrowing than has typically been done so far". Agreed; cf. my comments in a similar vein, 1991:15.

3. *Comments on "Exploring the Nostratic Hypothesis", §3 (and IE *mēms- 'meat, flesh')*

3.1. *Indo-European and Nostratic: A realistic approach*

At the beginning of their §3, AMR et al. make a valid criticism of my own critique of some of Illič-Svityč's etymologies: although I claimed that there were "many cases in which the conjectured root meaning and the meaning of the IE. forms show a troubling degree of semantic latitude" (1991:13), and that certain other sorts of etymological problems were "pervasive" (1991:14), I did no more than provide what I felt to be representative examples of each type, without actually documenting the full extent of these problems (by providing percentages, or any other statistical breakdowns based on the entire corpus of Illič-Svityč's Nostratic etymologies). In partial exculpation, I might note that the original version of my manuscript ran to 59 pages; for reasons of space, it proved necessary to omit discussions of additional examples beyond the skeleton crew of representative ones that survived into the published version. My critique, then, amounted to a "Where there's smoke, there's fire" argument, as opposed to a rigorous statistical demonstration, the absence of which AMR et al. properly register.

There was, however, another rationale behind my failure to provide a detailed statistical profile incorporating evaluations of each of Illič-Svityč's 378 Nostratic etymologies, or at least that substantial subset of them that incorporated Indo-European data: while I felt that I could form a reasonably accurate impression of the handling of Indo-European material by reading through the relevant lemmata, the expenditure of time and effort in performing the detailed evaluations that would have been required in order for me to publish responsible statistics went far beyond what I, as a working Indo-Europeanist/Classicist — an interested onlooker of Nostratic studies, but not a Nostraticist as such — was willing and able to devote. My preliminary

[16] See esp. Nichols and Peterson 1996:350ff. on nasals in pronominal roots, in particular "correlations of nasals with person categories" (351ff.), "phonosymbolism and other distributional universals" (354ff.), and "first person singular *m* in the Old World" (360).

efforts, then, were intended to suggest that from the vantage point of Indo-European linguistics a quarter of a century after Illič-Svityč's compilation, his handling of the Indo-European material was not as sacrosanct as seems to have been assumed by many of his followers. This point, in turn, is directly related to a more general one concerning the pursuit of Nostratic research. If it is a worthwhile enterprise to reevaluate the corpus of Nostratic etymologies with all of the rigor that modern historical linguistics can muster, it seems important to consider the most efficient manner in which such a task could be carried out. Without intending to impugn the talents and training of gifted polymaths like Illič-Svityč himself (not to mention many of his colleagues and followers, as well as other practitioners in the field), I believe that it is no longer realistic for a single scholar to control the data and the theoretical frameworks of all of the six "classical" Nostratic language families (to say nothing of additional ones) with sufficient competence and skill to be able to pursue such a full-scale evaluation in a scientifically responsible manner. If such a thing was marginally possible for Illič-Svityč, it is in my view no longer possible today. (In Indo-European alone, significant theoretical advances since Illič-Svityč's time cover all areas of the protolanguage from phonology to syntax; these have arisen, in part, from extensive revisions in our understanding of developments in certain language areas, such as Anatolian and Tocharian, as well as important new textual material, as in Celtic and Italic.) It seems to me, then, that future Nostratic research should begin to look toward cooperative and collaborative models, somewhat as in the hard sciences (and, indeed, as in the contribution by AMR et al.): that is, scholars with special expertise in individual language groups (but with a broad knowledge of Nostratic theory, and ideally some basic training in at least some of the other language families in question) could pool their efforts in evaluative projects, whether these are conceived on a grand or a modest scale. The grander collaborative projects, of course, might require major funding, which is admittedly scarce in the present political and academic climate. Nevertheless, this appears to me the most promising strategy for promoting the possibility of significant advances in the field of Nostratic.

3.2 *On IE *mēms- 'meat, flesh' and PN *Homsa 'id.'*

In noting that my comments about Illič-Svityč's handling of Indo-European in Nostratic etymologies "contain valid criticisms of particular elements in Illič-Svityč's work", AMR et al. reasonably observe that such criticisms "represent no more than points on which the theory would need revision". Indeed, they suggest that some of my objections (and, in principle, others like them) "could be a very useful guide to the cleaning up of Nostratic", and might "ultimately

lead to revisions that would strengthen [Nostratic theory]". AMR et al. go on to discuss two cases of this general sort: the first, which specifically addresses one of my critiques of Illič-Svityč, and directly involves the analysis of Indo-European material, will be discussed at some length in this section; the second, which is primarily concerned with Altaic, and in which Indo-European material figures only tangentially, will call for only the briefest comment in §3.3.

I concluded the analytical portion of my earlier paper with a discussion of Illič-Svityč's etymology for a Nostratic etymon meaning 'meat, flesh', reconstructed by him in the form PN *Homsa (1971:252f., #114) on the basis of material from Uralic and Indo-European (together with material Illič-Svityč labeled as questionable from Afroasiatic and Dravidian). Beside Uralic *omśa, Illič-Svityč set up the Indo-European form as "*(H)mēms-" (despite the lack of any evidence for initial laryngeal); in order to align the Uralic and Indo-European forms, he further postulated a contamination of (purely hypothetical) state I and state II forms "*$h_3$ems-" and "*$h_3$mes-", respectively, resulting in IE "*(h_3)mems-". As any practicing historical linguist knows, contaminations do occur under certain circumstances; but the available evidence renders this a totally unsatisfactory analysis, and I interpret AMR et al.'s wish to replace it as implicit acquiescence in that judgment.[17] In an attempt to account more satisfactorily for the IE form (including its unusual root shape and the pervasive lengthened grade), AMR et al. propose to explain the IE form as "some kind of complex form, perhaps a reduplicated *hme-hms- or a compound *(h)me-hms-", with the second member a regular zero-grade reflex of PN *Homsa. They find confirmation of this approach in a recent suggestion of Puhvel (1992:268f.), according to which Hitt. *hašš-* 'beget' and various other Anatolian forms referring to descendants (including CLuv. *hamša/i-* 'grandson') continue a PIE root *Hém-s-/*Hm-és-; Puhvel then compares this root directly with the IE 'flesh' word, which he takes to continue a reduplicated root noun *Hme-H(m)s-.

It would indeed be gratifying if my criticism of Illič-Svityč's reconstruction for the IE 'flesh' word led, as AMR et al. suggest, to a "refinement of [the Nostratic hypothesis], while clarifying a problematic area of Indo-European". But I regret that I cannot accept such credit as might accrue for such results, since the new proposal seems to me almost as unsatisfactory as that of Illič-

[17] Equally unsatisfactory, in my view, are "inner-Indo-European" attempts to connect *mēms- with the word for 'raw' (*(H)ōmó- or (H)oHmó- or (H)e$h_3$mó-; Ved. *āmá-* etc.) via an irregularly-reduplicated derivative with *-s- somehow related to Ved. *āmiṣ-* 'raw meat' (see Trubačev 1993:10 for references and discussion).

Svityč. At most, I would grant the possibility that Nostr. *Homsa 'meat, flesh' could be related to such forms as CLuv. *hamša/i-* 'grandchild', HLuv. *ha-ma-sa-* 'id.', Lyc. *xahba-* 'id.' (< Proto-Anatolian *Hamsa-),[18] presumably via some such semantic route as might be provided by expressions like Eng. 'flesh of my flesh' = 'offspring'.[19] It is impossible for me to accept, however, the further connection of this material with the IE word for 'flesh', in view of the following phonological and morphological problems.

Consider first the laryngeal color of the medial laryngeal in AMR et al.'s reconstruction *hme-hms-, under the assumption that this formulation involves reduplication. Given the lengthened e-grade of the IE 'flesh' word, one must in fact specify this medial laryngeal as *h_1: ignoring, for the time being, the question of the initial laryngeal, medial *h_2 would yield (late) IE **māms- (not *mēms-), and medial *h_3 would yield (late) IE **mōms- (not *mēms-). One could salvage a reconstruction with *h_2 or *h_3 only by recourse to "Eichner's Law", according to which coloring would not take place with a preceding lengthened-grade vowel (thus, for *h_2, *$\bar{e}h_2$ > /ē/, not /ā/)[20]; such a ploy, however, is hardly acceptable, since this would require lengthened grade in a reduplicating syllable (i.e. *(h_2)mē-$h_2$ms- or *(h_3)mē-$h_3$ms-), an utterly anomalous configuration. IE *mēms-, then, entails a reduplicated *(h_1)me-$h_1$ms- (under this theory); but *(h_1)me-$h_1$ms- is incompatible with Anatolian, where the initial laryngeal of CLuv. *hamša/i-* etc. requires IE *h_2 or *h_3. This conundrum, then, appears to me to be fatal for AMR et al.'s "reduplicated *hme-hms-", at least in terms of the comparison of the IE 'flesh' word with the Anatolian forms in question.[21] AMR et al. seem to offer, however, a second option, i.e. "a compound *(h)me-hms-". The implications of this designation, and of its accompanying parenthetical notation, are not specified; one suspects, nevertheless, that the use of the term "compound" is designed to help justify the absence of any IE reflex of initial laryngeal, as might otherwise (under

[18] But NOT (despite AMR et al. and Puhvel) to Hitt. *hašš-* 'beget' and its derivatives, since these forms continue Proto-Anatolian *Hans-, not *Hams-; see Szemerényi 1987:354 and especially Melchert 1994:164f.

[19] For further etymological interpretations of CLuv. *hamša/i-* etc. in terms of an IE *$h_2$em-so- (with a possible Greek comparandum), see Bader 1988:33f.

[20] For details, see e.g. Mayrhofer 1986:132f., 141f.

[21] If the vocalic initial of the Uralic and Dravidian forms (Illič-Svityč is less clear about the probative value of the Afroasiatic material) is likewise compatible only with IE *h_1, then even the limited comparison between PN *Homsa 'flesh' (where H may call for IE *h_1) and Anatol. *Hamsa- 'offspring' (where H here calls for IE *h_2 or *h_3) invites considerable caution. On the other hand, if the Uralic o-vocalism is taken to require IE *h_3 (as Illič-Svityč evidently assumed), this again produces difficulties (see n. 23 below for details).

most versions of IE laryngeal theory) be expected to surface in various ways, such as prothetic vowel in Greek and Armenian (lacking: Gk. μηρός, Arm. *mis*), or pre-laryngeal compound lengthening in Indo-Iranian (no evidence available for this word, since no old compounds of the requisite format are attested).[22] Thus, if "*(h)me-hms-" is a "compound", one could then appeal to the phenomenon of laryngeal loss in compounds (itself, it should be noted, not uncontroversial: see Lindeman 1990 for recent discussion) in order to justify the apparent absence of the initial laryngeal, since loss of laryngeal in a reduplicating syllable is not otherwise known (cf. "Attic reduplication", as in ἐλήλυθα < *$h_1$le-$h_1$ludh-, not **le-$h_1$ludh-). But if one takes these assumptions seriously, one arrives at the attendant claim that "*(h)me-" (i.e. *(h_2)me- ? *(h_3)me- ?) is in effect a *different root* from that which appears in "*-hms-" (i.e. *-$h_1$ms-) — hardly an attractive result, and one which raises at least as many questions as it would answer.[23] An additional phonological objection concerns expectations about the syllabification of a sequence like */-meHms-/: to judge, for example, from IE *meh$_1$ns- 'moon; month' and *$h_2$weh$_1$-nt-o- 'wind', the resonant after the laryngeal should be treated as syllabic in Indo-Iranian, at least vestigially (cf. Vine 1990, with n. 17). The Rig-Vedic and Avestan material, though scanty, shows no such treatment, pointing instead to a simplification /mēs-/ < /mēms-/.[24] These facts suggest, then, that a medial laryngeal is not in fact to be reconstructed, and that the form instead shows a non-laryngeal lengthened grade. Indeed, this is further suggested by Toch. B *mīsa* 'meat' (plurale tantum,[25] from the collective of the IE root noun, cf. Ved. *mā́s*): as recently discussed by Ringe (1996:70f.), the root vowel of *mīsa* cannot continue PIE *ē, but seems to show rather *e, pointing therefore to an acrostatic root noun *mḗms-/*méms- underlying the

[22] The absence of evidence for initial laryngeal was one of the points on which I had insisted in my original critique of Illič-Svityč's proposal (1991:31).

[23] At best, one could salvage the etymology by postulating a scenario like the following:
(i) a reduplicated root noun PIE *$h_3$me-$h_3$ms- (see below on the morphology) > *me-$h_3$ms- (with irregular and extremely early loss of laryngeal in reduplicating syllable);
(ii) the resulting *me-$h_3$ms-, no longer felt to be reduplicated, is reanalyzed as *meh$_3$ms- (> IE **mōms-, nowhere attested);
(iii) *meh$_3$ms- is for some reason replaced by lengthened-grade *mēh$_3$ms-.
Apart from the obvious weaknesses of such an account, see the immediately following discussion for further problems.

[24] On the formulation of this rule see Mayrhofer 1986:163n277 and 1994:344, with references.

[25] An archaic feature for this word, on which see Degtjarev 1993.

IE word.[26] A final phonological factor to be weighed in this connection is the testimony of the (Balto-)Slavic intonation: if, following Kortlandt (1985, 1988), lengthened grade vowels regularly show circumflex intonation in Balto-Slavic, then the available evidence (e.g. SCr. *mêso*) once again points to lengthened grade, as opposed to laryngeal lengthening.[27]

On the morphological side: Puhvel's conception of a "reduplicated root noun" *Hme-H(m)s-, which AMR et al. appear to adopt without question, involves a construct of the most dubious sort (even apart from the morphological implications of Toch. B *mīsa* and the Balto-Slavic intonation). To begin with, note that the form of the reduplication itself poses a problem. In general, "full" reduplication of this sort is found (mainly in certain verbal categories) with roots of the shape *HReC- (see Strunk 1988 for discussion); but there is again no evidence for a hypothetical "state II" **Hmes-, which brings us perilously close to one of the signal weaknesses of Illič-Svityč's contamination theory. The presumed morphological category of "reduplicated root nouns" is also problematic. As shown by Hamp (1972), IE nouns with e-reduplication were regularly derived from verbal bases (thus *k^we-k^w]-o- 'wheel' to *k^wel- 'turn', etc.); but no such verbal base is quotable for 'flesh'.[28] Moreover, IE nouns with e-reduplication are normally thematic (as in 'wheel'), or (more rarely) appear as i- or u-stems[29]; an apparent root-noun form of the exceedingly rare type Hitt. *memal* 'meal' (NB: with full grade!) probably belongs rather with the intensive type (i.e. normally with intensive reduplication) seen in forms like Ved. *jógū-* 'loud-singing' or Gk. λαῖλαψ 'storm' (thus Schindler 1972:8).[30] There is thus nothing about an alleged "reduplicated root noun *Hme-H(m)s-" that would fit very well with — much less clarify — the structure of IE nouns with e-reduplication.[31]

[26] For the terminology used here ("acrostatic" [Germ. *akrostatisch, akrodynamisch*] = root accent in both strong and weak case forms) see e.g. Meier-Brügger 1992:II.71, following Rix 1976:123.

[27] Kortlandt (e.g. 1988:301) cites Li. *mėsà* as an example of lengthened grade vowel in an original root noun, but the status of the Baltic material for the 'flesh' word is somewhat unclear, given the possibility that some forms have been borrowed from Slavic (see Trubačev 1993:9 for discussion and references).

[28] Recall that one cannot appeal to a connection with a verbal root meaning 'beget' or the like, as in Hitt. *hašš-* (cf. n.18 above).

[29] See Hamp's discussions of OIr. *feóil* 'flesh, meat' (i-stem) and the various words for 'beaver' (contaminations involving o-stem and u-stem forms).

[30] For certain additional features specific to *memal* (together with possible Armenian comparanda) see Joseph 1992.

[31] I cannot accept Puhvel's claim (loc. cit.) that zero-grade *HmḪ(m)s- appears in the *mas-* of Lat. *masculus* 'male (adj.)' and the non-nominative forms of *mās* - 'male' (gen. *maris*), but full discussion would be otiose here (among other problems, note that such an account has no way of accounting for the m-loss, at least according to the usual formulation of the rule [cf. n.24 above], and the same applies, moreover, to Puhvel's "*Hme-H(m)s-"); for a different conception of this material, see Schrijver 1991:167f.

For all of these reasons, then, this particular case cannot in my opinion serve as an example of a revision that would strengthen Nostratic theory. This does not, of course, exclude the possibility that such revisions could be found, and does not of itself invalidate the Nostratic theory.

3.3 *On sibilant clusters and affricates*

The adjustment AMR et al. propose to Illič-Svityč's system is surely attractive, on general phonological grounds. One imagines, however, that many details remain to be worked out. In the example discussed: if the /sk-/ of IE *(s)kerb(h)- reflects the Nostratic initial cluster more or less accurately, as AMR et al. seem to assume, it seems slightly odd that such a cluster would simplify, in Japanese, to /t-/ (as opposed to, say, /k(k)-/, as in Middle Indic). Since, on the other hand, the simplification of an affricate /č-/ (i.e. [tʃ]) to /t-/ is exceedingly natural, one would like to know why Japanese /t-/ is assumed to reflect a simplification of the original Nostratic cluster /sk-/, and not a simplification of its resulting Altaic affricate /č-/. But this is a point of detail that does not weaken AMR et al.'s general claim about this example.

4. *Comments on "Exploring the Nostratic Hypothesis", §4 (and IE *pénk^{w}e 'five')*

4.1 *Introductory*

In the fourth section of their paper, AMR et al. point to various "revisions and refinements that further work [in the daughter languages] has begun to contribute to the Nostratic hypothesis", including "some new etymologies that, in a simple and convincing way, cast light on well-known mysteries within one or another of the six subfamilies". Surely there can be no quarrel with what I take to be the general point of this section, namely that advances in our understanding of the various daughter languages can be expected to contribute to the development of Nostratic theory. Indeed, it was partly the sheer bulk of the advances in certain of the daughter languages that led me to suggest (cf. §3.1 above) the need for cooperative research.

Among the "new discoveries which lend confirmation to the overall theory of Nostratic linguistic relatedness", AMR et al. devote their most extended discussion to what they term an "intricate" and "compelling" new etymology involving IE *pénk^{w}e '5'. The material is intricate, but in a sense, not intricate

enough: there are, to my mind, too many unanswered questions to allow me to concur with the evaluation that this etymology is "compelling". I do, however, find the hypotheses to be interesting and potentially important. The brief assessment that follows is therefore offered partly to justify my skepticism, but partly also in the hope that it may help to refine these hypotheses further.

4.2 *Notes on IE *pénk^{w}e '5' and related (?) forms*

It must be stated at the outset that the etymological literature on the IE numerals is immense, and that the present discussion in no way attempts to be exhaustive even concerning all of the issues surrounding '5'. For the most part, the following discussion is restricted to comments directly related to AMR et al.'s treatment of IE *pénk^{w}e '5' and certain forms taken to be related to it.

A preliminary point concerns the Greek and Sanskrit material AMR et al. discuss towards the beginning of their presentation.[32] For Gk. πεμπάζομαι/ πεμπάζω 'count',[33] πεντάχα·ἡ χείρ Hsch., Tsak. πεντόχτη 'hand', and Classical Sanskrit *prapañcayati* 'develop, amplify, treat at length', AMR et al. first note the standard assumption that such words are based on words for '5', but then observe: "Such a direction of semantic evolution is quite unusual compared to the development of numerals in other language families, where the direction always seems to be from body part term to numeral". They thus conclude: "It seems more likely that the primary meaning was not "five" but rather something to do with the fingers." If the phrase "primary meaning" refers to IE *pénk^{w}e, there is no particular problem: it might indeed be productive, as AMR et al. go on to suggest, to reconsider the possibility that the IE word for '5' actually derives from a word referring somehow to the hand. But if the phrase "primary meaning" refers to the Greek and Sanskrit material discussed directly above this statement, the conclusion is unacceptable. As rare as the reverse development may be, it certainly occurs (one need think only of English expressions like 'high five', 'Slap me five!', etc., where 'five' = 'hand'[34]); and in any case, the derivation of this Greek and Sanskrit material from words for '5' is beyond question, given the nature and date of the attestations, and the structure of the formations themselves (thus πεμπάζομαι [stem πεμπαδ-] clearly based on πεμπάς '5; group of 5' [stem πεμπαδ-], etc.). Thus Mayrhofer (1992:66) was clearly correct in his judgment that Skt. *prapañcayati* is "ohne Bedeutung für die Herkunft von *pañca* usw.", and the same applies to the Greek forms. This material, then, has no bearing whatsoever on the question of the original meaning of IE *pénk^{w}e.

The main part of AMR et al.'s discussion concerns the possible cognacy of their Nostratic proto-form *p'ayngV (based on Uralic and Altaic evidence, and with meaning roughly "some way of holding the hand which almost certainly involved holding all five fingers together") and PIE *pénk^{w}e '5'. According to AMR et al., the two main problems in aligning these forms involve the IE labiovelar, and the nature of its phonation type. For the first, AMR et al. offer a rationale that would allow them to reconstruct a final *-o in the Nostratic form, which would in turn yield an IE labiovelar regularly. As for the second, AMR et al. argue (following Illič-Svityč and others) that since the predicted IE **peng^{w}h- would violate IE root-structure constraints, such a form would have been replaced by a phonotactically well-formed variant, i.e. *penk^{w}- or *peng^{w}-.

It would certainly be interesting to be able to align in this way the IE word for '5' with a word for (some way of holding the) hand reconstructable from Uralic and Altaic. To my mind, however, the most difficult and puzzling IE material has not thereby been explained. If, for example, PN *p'ayngV (or, as AMR et al. prefer, *p'ayngo) designated the hand or some way of holding it, and if IE *pénk^{w}e continues the same word, how exactly are we to interpret the semantic and formal development of the IE form? Was this simply a straight development 'hand' > 'five', with no accompanying derivational apparatus?[35] If so, what is the final *-e in the IE form? Why, moreover, is '5' undeclined in IE, when (a) other IE nouns traceable to Nostratic etyma are fitted out with normal IE declensional paradigms, and when (b) even the preceding four numerals themselves were declined in IE? Finally, what is the precise nature of the relationship between *pénk^{w}e and certain Balto-Slavic and Germanic words often taken to be related to it with meanings 'fist' and 'finger' (cited by AMR et al. at the beginning of their discussion, but not otherwise incorporated into their account)? These latter forms, I suggest, may be of central importance for the entire problem. The intricacy of the issues involved is such that a full treatment cannot possibly be offered here, as I have already noted; the following sketch merely offers, *exempli gratia*, one possible scenario that may help justify my assertion concerning the Balto-Slavic and Germanic material, as well as the level of my skepticism regarding the proposed PN *p'ayngV (*p'ayngo ?) : PIE *pénk^{w}e etymology.

[32] AMR et al. mention the Germanic word for 'finger' (as interpreted etymologically by Meillet and Watkins) in the same context, but I reserve this form for discussion below.

[33] First attested as a hapax in the *Odyssey* (aor. subj. πεμπάσσεται, δ 412).

[34] See also e.g. Chapman 1986:136, s.v. five, meaning 2 ('*n jive talk* The hand; the five fingers').

[35] AMR et al. invoke Uto-Aztecan for this semantic development, but one would like to see the derivational details of comparable patterns.

For the sake of argument, let us begin by taking at face value the Altaic evidence bearing on the final vowel, leading to a PN *p'aynga (not *p'ayngo). Let us then make the following three uncontroversial assumptions:

(i) a PN root-skeleton *p'–g would yield not the phonotactically ill-formed IE **p–gh, but rather IE *p–k or *p–g (similarly AMR et al., as noted above);
(ii) a PN sequence *-ga yields an IE pure velar;
(iii) PN unstressed *-a normally disappears in IE (explicitly e.g. Dolgopolsky 1992b:304).

In this way, PN *p'aynga can regularly yield a PIE root *penk-. Note further that the Balto-Slavic 'fist' word and the Germanic word for 'finger' are most straightforwardly reconstructed on the basis of such a form (i.e. *pn̥k-sti- for the former, and *penk-ró- for the latter); the traditional reconstructions that operate with labiovelar are purely conjectural, and based on the assumption of a connection with '5'.[36] Indeed, the fact that IE *penk^{w}- underwent assimilation in Germanic to *pemp- (hence Goth. *fimf* etc.) speaks against assigning *pn̥k$^{(w?)}$-sti- and *penk$^{(w?)}$-ró- directly to the word for '5'.[37]

A further argument in favor of reconstructing an IE *penk- meaning roughly 'hand' (thus already Szemerényi and others, see the references at Mayrhofer 1992:66) arises from the morphology of the word for 'finger'.[38] IE nominal *-ro- (with both full-grade and zero-grade root) has so far received little attention, as opposed to the familiar adjectival *-ro- (generally *-ró-, i.e. with zero-grade root).[39] Although this does not seem to have been noted, part of the profile of nominal *-ro- includes forms showing a function that, depending on the nature of the root, could be described variously as "partitive" or "singulative" or "individuating". Thus, as already discussed, IE *mēms- 'flesh' is reconstructable as a root noun, partly attested in plural forms ultimately

[36] Despite the handbooks (e.g. Pokorny 1959:808 s.v. *penkᵘe*, ad fin.), there is no necessary connection between Gmc. 'finger' and the ordinal forms of '5' with r-formant, as in Armenian and Celtic.

[37] It is even possible to reconstruct the 'fist' word as *pn̥k-sti-, as A. Dybo prefers to do (1996: 83), on the basis of Li. *kùmstė* (thus with recourse to a contact dissimilation *pn̥k-sti- > *kn̥k-sti-, vs. the metathesis (Ba.) *punk- > *kump- [cf. Li. *kepù* beside OCS *pekǫ* 'cook'] normally assumed). But in this case, the *-n-* of OPr. *kuntis* is exceedingly difficult to account for (see the discussion in Toporov 1984:291).

[38] The formation of the 'fist' word is quite obscure, although its general similarity with forms like Ved. *muṣṭí-* 'fist' has long been noted (see e.g. Mayrhofer 1994:363, as well as 1996:647 [s.v. *śúpti-* 'shoulder'] on IE *-ti- in body part terms).

[39] I hope to take up this problem in more detail elsewhere.

derived from the collective. But a nominal *mē(m)s-ró- is also clearly reconstructable for IE (Lat. *membrum* 'limb', Gk. μηρός 'thigh, thigh-piece', OIr. *mír* 'portion [of meat], morsel [of food]'), and is perhaps to be interpreted, on the basis of the meanings of its reflexes, as having originally meant something like '[individual] *piece of* flesh' (as opposed to 'flesh' more generally or collectively).[40] Similarly, nominal *-ro- forms built to verb roots often express instrument, result or other concrete notions — i.e., so to speak, individual instantiations of the action of the verb: thus *deh$_3$-ro- 'gift' (to *deh$_3$- 'give') in Gk. δῶρον, OCS *darŭ;* *h$_2$eǵ-ro- 'tract' (to *h$_2$eǵ- 'drive') in Gk. ἀγρός, Lat. *ager,* Skt. *ájra-* etc.; *neḱ-ro- 'corpse' (to *neḱ- 'die') in Gk. νεκρός; etc. Such formations are found with some frequency in Germanic and Balto-Slavic: e.g. *leǵh-ro- (to *leǵh- 'lie') in Goth. *ligrs,* OHG *legar* 'bed, couch' (i.e. '[individual] thing for lying'), *dem-ro- (to *dem- 'build') in OIc. *timbr,* OE *timber* etc. 'timber' (i.e. '[individual] piece of building material'), *nas-ro- (to *nas- 'nose') in OCS *nozdri* 'nostrils' (i.e. '[individual] nasal passages'), cf. Li. *nasraĩ* 'throat'; etc. Thus *penk-ró- would mean precisely '[individual] part of the hand', i.e. 'finger'.

If, then, PN *p'aynga 'hand (vel sim.)' has developed into an IE *penk- 'id.' (vestigially attested in Balto-Slavic and Germanic derived forms), there is no absolute necessity to compare these forms directly with the IE words for '5': in principle, the phonetic similarity between these forms could have arisen in various ways, including via folk-etymological adjustments no longer recoverable. If one did, however, wish to pursue more explicit means of connecting *penk- and *pénk^{w}e, one might well wish to reconsider the old idea that '5' actually ends with the familiar enclitic connective *-k^{w}e 'and'. For example: one could imagine a primordial counting-sequence '1, 2, 3, 4, and [full] hand', or '1, 2, 3, 4, and fist', in which, during the utterance, the words for the numerals would correspond with finger-counting motions; the phonological word 'and-hand' in this sequence (composed of 'hand' plus

[40] In addition to the reduplicated root noun discussed above, Puhvel also sets up (loc. cit.) a "neuter r-stem extension" *Hme-Hms-ṛ, the thematized version of which he takes to account for Lat. *membrum* etc.; but such a construct introduces a considerable degree of derivational complexity. A non-vacuous thematization should yield an essentially adjectival derivative ('fleshy' or the like), which would then need to have been substantivized to account for *mēms-r(-)o- '[piece of] flesh'. And while r/n-stems are occasionally found beside root nouns, one would not expect *both* formations to be reduplicated; a reduplicated r(/n)-stem would in any case be an undesirable morphological *unicum.* In contrast, the independent existence of IE "nominal *-ro-" renders this entire problematic apparatus unnecessary.

enclitic *-k^{w}e) would then have assumed the meaning 'five'.[41] A syntactic or phraseological conception of this sort might also help to account for other explananda, including why the series of declined numerals ends precisely with '4'. The resulting **pénk-k^{w}e would then have undergone reduction to *pénk^{w}e, a process which again could in principle be characterized in a variety of ways: thus, as Brian Joseph reminds me, one could think in terms of a resyllabification *pén$kk^{w}e, with subsequent simplification of the cluster, exactly as in Strunk's analysis of Gk. αἰπόλος 'goatherd' (1992; similarly Meier-Brügger 1992:I.92) < *ai$kk^{w}olos < *aik-k^{w}olos (with voicing assimilation from an original *aig-k^{w}olos).[42] A final complication concerns the possible existence of an IE *pen- 'hand',[43] which could have interacted in various ways with the material discussed above.

Whatever one thinks of the details of these lucubrations, they do, I hope, demonstrate the extent to which simply juxtaposing a reconstructed PN *p'ayngo with PIE *pénk^{w}e fails to address a series of relevant forms and problems, and why this etymology does not strike me as a "compelling" one that exemplifies how "Nostratic connections may shed light on problems of the individual branches that cannot be solved by considering data from only those branches". But the existence of a possible "Proto-Uralo-Altaic" *p'aynga or *p'ayngo is undoubtedly of potential interest for PIE *pénk^{w}e, and may ultimately, given a sufficiently detailed account, yield the sort of results AMR et al. have sought to illustrate.

5. *Conclusions*

AMR et al. conclude (§5) by noting that "the question ... is ... whether the evidence here is sufficient to justify further serious investigation" of Nostratic; they respond with an "overwhelming yes". Despite my criticism of various points of detail concerning the Indo-European material AMR et al. discuss, I hope it is clear that I share in their affirmative response. In view of those criticisms, however, I would continue to insist, as I did at the conclusion of my earlier paper (1991:32), that the eventual success any such serious investigation may find will be proportional to the precision and depth of its analyses, which (as I suggested above) would very likely be enhanced by structured programs of cooperative and collaborative research. If I have not found the requisite precision and depth in the particular analyses AMR et al. propose, I am not thereby deterred from looking forward to more satisfying attempts.

[41] The relatively early loss of a *penk- 'hand' (vel sim.) as an independent primary lexeme (vs. derived forms *pn̥k-sti- 'fist' and *penk-ró- 'finger') could have contributed to the development 'and-hand' > 'five'.

[42] Alternatively, if one accepts the Nostratic pedigree of IE *-k^{w}e itself (PN *ḳ/o/, Illič-Svityč 1971:325f., #201), one could even begin directly with a PN *p'aynga-ḳ/o/ 'and-hand' (or 'and-fist', etc.) > pre-PIE *péngh-k^{w}(e) 'five' (> PIE *pénk^{w}e), which might obviate the need to invoke what AMR et al. refer to as the "Procrustean bed of Indo-European root structure". But a *penk- still seems needed, in any event, if not for *pn̥k-sti- (conceivably derivable via assimilation from a *pn̥gh-sti-), then at least for *penk-ró-.

[43] References at Mayrhofer 1992:66, to which add Burrow 1971:542.

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